

AP Precalculus: The Complete Exam Guide

The three tested units explained, the calculator and no-calculator split, a modeling playbook, and 10 hard practice questions with full explanations.

Hi, I'm **Pigo!** I'll drop tips all through this guide.

**2:55**

hours total
105 + 70 minutes

42+4

multiple-choice +
free-response questions

3 units

tested on the exam
(Unit 4 is not)

81–82%

scored 3 or higher
in recent years

INSIDE THIS GUIDE

01	The exam at a glance	2	06	10 traps that cost points	7
02	Section I: multiple choice	3	07	Your study roadmap to May	8
03	The units, in plain English	4	08	10 practice questions	9
04	Section II: free response	5	09	Answer key & explanations	10
05	Anatomy of a full-credit answer	6	10	Scores & exam-day checklist	11

01 THE EXAM AT A GLANCE

Two sections, four parts

Your AP score (1–5) combines multiple choice (about 63%) and free response (about 37%). Both sections have a no-calculator part and a graphing-calculator part.

44%

MC Part A • 29 Q • no calc

19%

MC Part B • 13 Q • calc

19%

FR Part A • calc

19%

FR Part B • no calc

Exam structure

Section	Weight & timing	What you do
I • Multiple choice	≈63% of score 1 hr 45 min 42 questions	Part A: 29 questions in 65 minutes, no calculator (≈44%). Part B: 13 questions in 40 minutes, graphing calculator required for some (≈19%).
II • Free response	≈37% of score 1 hr 10 min 4 questions	Part A (calculator, 35 min): Q1 Function Concepts • Q2 Modeling a Non-Periodic Context. Part B (no calculator, 35 min): Q3 Modeling a Periodic Context • Q4 Symbolic Manipulations. Each is worth 6 points.

HYBRID DIGITAL

Click in Bluebook, write on paper

You answer multiple-choice questions and read the free-response questions in the **Bluebook** app, then handwrite your free-response answers in a paper booklet. Practice writing clear, organized work by hand.

WHAT'S TESTED

Units 1–3 only

The exam covers **polynomial and rational**, **exponential and logarithmic**, and **trigonometric and polar** functions. Unit 4 (parameters, vectors and matrices) is taught but not tested.

How to use this guide

Just starting?

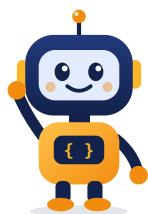
Read pages 2–6, then follow the roadmap on page 8 month by month.

Mid-year?

Check the traps on page 7 and drill the free-response playbook on page 6.

Exam in weeks?

Take the 10 questions on page 9 in 22 minutes, then use the final-four-weeks plan.



Pigo's tip: Precalculus is about **representations**: the same function as a graph, table, equation and verbal description. Practice switching between all four.

SECTION I

02 The multiple-choice section

Part A gives you 29 questions in 65 minutes without a calculator (about 2¼ minutes each). Part B gives 13 questions in 40 minutes with a graphing calculator (about 3 minutes each).

29

no-calculator questions in Part A

13

calculator questions in Part B

≈63%

of your AP score

0

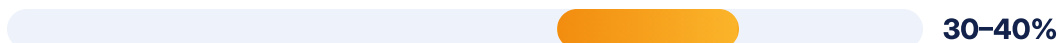
points lost for a wrong answer, so never leave one blank

How the questions are weighted by unit

College Board gives a range for each unit's share of the multiple-choice section. Bars show minimum to maximum.

Unit 1

Polynomial & Rational Functions



30–40%

Unit 2

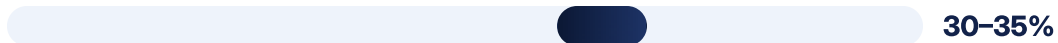
Exponential & Logarithmic Functions



27–40%

Unit 3

Trigonometric & Polar Functions



30–35%

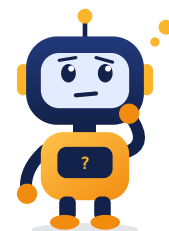
0% 10% 20% 30% 40% 50%

What this means for your study time: The three units are close to equal. Your weakest unit is your biggest opportunity, and trigonometry is where most students lose points.

Strategies that work on this section

- **Use rates of change.** Constant first differences mean linear; constant second differences mean quadratic; constant ratios mean exponential.
- **Factor first** to find zeros, holes and vertical asymptotes of rational functions.
- **Log rules are tools.** Combine logs before solving, then check that answers are in the domain.
- **Know the unit circle** cold for Part A: exact values at multiples of $\pi/6$ and $\pi/4$.
- **In Part B**, graph and use intersect or zero instead of solving by hand.
- **Flag and move on** after 3 minutes. Return once the quick points are banked.

Part A is **no calculator**. Unit-circle values and log rules need to be automatic.



03 WHAT THE COURSE COVERS

The 4 units, in plain English

Units 1–3 are tested; Unit 4 is taught but not on the exam.

UNIT

1

30–40%

Polynomial & Rational Functions

How functions change, and the shapes of polynomials and rational functions.

Be ready to: describe rates of change and concavity; find zeros with multiplicity and end behavior; identify holes, vertical and horizontal asymptotes; solve inequalities; work with transformations and function models.

UNIT

2

27–40%

Exponential & Logarithmic Functions

Growth and decay, and the inverse functions that undo them.

Be ready to: compare arithmetic and geometric sequences; build exponential models; use log properties to solve equations; find inverses and compositions; use semi-log plots to linearize data.

UNIT

3

30–35%

Trigonometric & Polar Functions

Periodic behavior, the unit circle, and polar graphs.

Be ready to: use sine, cosine and tangent on the unit circle; find amplitude, period, phase shift and midline; model periodic situations; use identities and inverse trig to solve equations; graph polar functions and analyze rates of change of r .

UNIT

4

Not tested

Parameters, Vectors & Matrices

Taught in the course to prepare for calculus, but not on the AP Exam.

Be ready to: use parametric functions, vectors and matrices for transformations. Your teacher may still assess it in class.

Looking ahead to calculus? Rates of change, function composition and inverses in this course are the exact ideas AP Calculus builds on.

04 SECTION II The 4 free-response questions

70 minutes, 4 questions, 6 points each. Part A (calculator) gets 35 minutes; Part B (no calculator) gets 35 minutes.

Q1 • 6 pts • calc

Q2 • 6 pts • calc

Q3 • 6 pts

Q4 • 6 pts

24 raw points in total, worth about 37% of your AP score

Q1

Function Concepts

Part A • calculator

Given a graph or table, describe behavior: where the function increases, its rate of change, concavity, and values of compositions or inverses.

Use precise words: "increasing at an increasing rate".

Q2

Modeling a Non-Periodic Context

Part A • calculator

Build a linear, quadratic, exponential or logarithmic model from data, use it to predict, and explain its limitations.

Write the model with numbers, then interpret each parameter in context.

Q3

Modeling a Periodic Context

Part B • no calculator

Build a sinusoidal model from a situation such as a Ferris wheel or tides, then interpret amplitude, period and midline.

Find the midline and amplitude from the max and min first.

Q4

Symbolic Manipulations

Part B • no calculator

Rewrite and solve expressions and equations using exponent, log and trig identities.

Show each algebra step; points are given for correct intermediate forms.

◆ A time plan for the 70 minutes

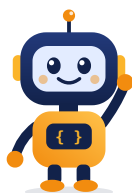
Q1
~17 min

Q2
~18 min

Q3
~17 min

Q4
~18 min

Part A and Part B are timed separately, 35 minutes each.



On modeling questions, **interpret in context**: "The population grows by 8% per year," not just " $b = 1.08$ ".

FREE-RESPONSE PLAYBOOK

05

Anatomy of a full-credit answer

Modeling questions reward clear setup and interpretation. Here is a short practice prompt with a full-credit answer.

PRACTICE TASK • WRITTEN BY ACT SAT GO

Practice prompt (Q3 style): A Ferris wheel has a diameter of 18 m, and its lowest point is 3 m above the ground. It completes one full rotation every 30 minutes. A rider starts at the lowest point at $t = 0$. (a) Write a function $h(t)$ for the rider's height. (b) Find the first time the rider is 16.5 m high.

Full-credit response

- ① Midline = $3 + 9 = 12$ m. Amplitude = $18/2 = 9$ m.
- ② Period 30 min, so $b = 2\pi/30 = \pi/15$.
- ③ Starting at the minimum: $h(t) = 12 - 9 \cos(\pi t/15)$.
- ④ $16.5 = 12 - 9 \cos(\pi t/15) \rightarrow \cos(\pi t/15) = -1/2 \rightarrow \pi t/15 = 2\pi/3$.
- ⑤ $t = 10$ minutes after the ride starts.

- ① **Midline and amplitude first**, from the minimum and maximum heights.
- ② **Period to frequency**: $b = 2\pi \div \text{period}$.
- ③ **Choose $-\cos$** because the rider starts at the minimum.
- ④ **Use exact unit-circle values** without a calculator.
- ⑤ **Answer in context** with units.

Five habits for every free-response question

1

Label every part: (a), (b), (c).

2

Write models with numbers, not just letters.

3

Interpret parameters in context with units.

4

Keep exact values in Part B (π , $\sqrt{3}$, fractions).

5

Stuck? Write the general form; it can still earn a point.

06 LEARN FROM OTHERS

10 traps that cost points

Meet **Glitch**. He lives in these ten mistakes. Squash each one before May.

Log of a negative? I love it!



01 Holes vs asymptotes

A factor that cancels gives a hole, not a vertical asymptote. Factor before deciding.

02 Extraneous log solutions

After solving log equations, check every answer: arguments of logs must be positive.

03 Percent growth over different periods

$Q = a(1.2)^{t/3}$ grows 20% every 3 years, which is about 6.3% per year, not 20%.

04 Degrees vs radians

AP Precalculus uses radians. Check your calculator mode before Part B.

05 Period vs frequency

For $\sin(bx)$, the period is $2\pi/b$. A bigger b means a shorter period.

06 Multiplicity ignored

At a zero of even multiplicity the graph touches and turns; odd multiplicity crosses.

07 Losing inverse-trig range

$\arcsin$ and $\arctan$ give values in $[-\pi/2, \pi/2]$; $\arccos$ gives $[0, \pi]$. Find the other solutions yourself.

08 Wrong end behavior

End behavior depends on the leading term only, including its sign and degree.

09 Negative r in polar

When $r < 0$, the point is plotted in the opposite direction of θ .

10 Bare numbers in context

Modeling answers need units and a sentence about what the number means.

Quick self-check: $f(x) = (x^2 - 9)/(x^2 + x - 6)$. Can you name the hole, the vertical asymptote and the horizontal asymptote? (Answers: hole at $x = -3$, VA $x = 2$, HA $y = 1$.)

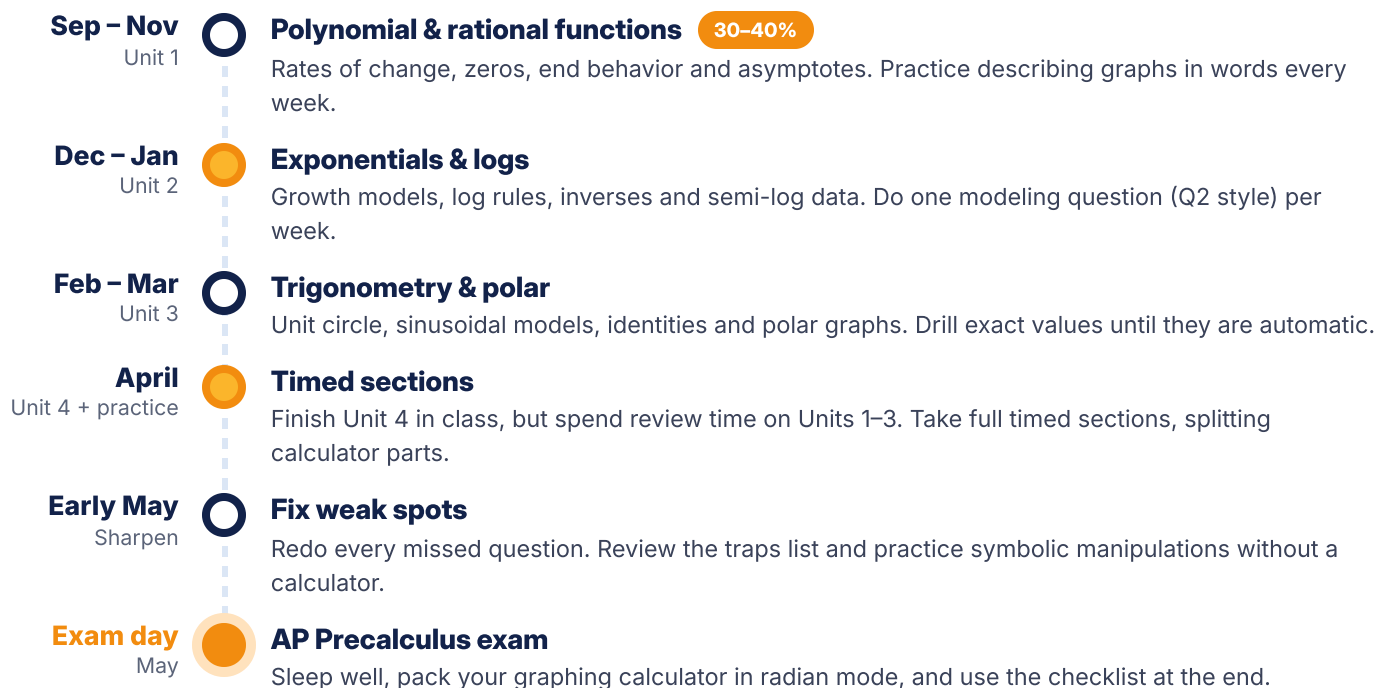
07

PLAN YOUR YEAR

Your study roadmap to May



A steady plan beats a last-minute rush. Shift the months to match your teacher's pacing, but keep reviewing old units as you learn new ones.



◆ The final four weeks

Week	Focus	Target
4 weeks out	Diagnose: one full Part A and two free-response questions	List your weakest unit
3 weeks out	Drill weak unit; one free-response question of each type	80%+ on unit drills
2 weeks out	Full practice exam under timing	Finish every part on time
Final week	Light review: unit circle, log rules, asymptotes	Rest the night before

08 TEST YOURSELF 10 practice questions

Original questions written by ACT SAT GO in the style of the exam, pitched at the harder end. Give yourself **22 minutes** for all ten, then check page 10.

1 UNIT 1 • RATIONAL FUNCTIONS

For $f(x) = (x^2 - 4)/(x^2 - x - 2)$, which statement is true?

- A f has a hole at $x = 2$ and a vertical asymptote at $x = -1$
- B f has vertical asymptotes at $x = 2$ and $x = -1$
- C f has a horizontal asymptote at $y = 4$
- D f has a hole at $x = -1$

2 UNIT 1 • POLYNOMIALS

For $p(x) = -2x(x - 3)^2(x + 1)^3$, on which interval(s) is $p(x) > 0$?

- A $(-\infty, -1)$ and $(0, 3)$
- B $(0, 3)$ and $(3, \infty)$
- C $(-1, 0)$ and $(3, \infty)$
- D $(-1, 0)$ only

3 UNIT 2 • EXPONENTIAL MODELS

A quantity is modeled by $Q(t) = 50(1.5)^{t/5}$, with t in years. By about what percent does Q increase each year?
(Calculator)

- A 8.1%
- B 8.4%
- C 10%
- D 50%

4 UNIT 2 • LOG EQUATIONS

What is the solution set of $\log_2(x) + \log_2(x - 6) = 4$?

- A $\{8\}$
- B $\{-2, 8\}$
- C $\{4\}$
- D $\{10\}$

5 UNIT 2 • INVERSES

If $f(x) = 3 \cdot 2^{x-1} + 4$, what is $f^{-1}(28)$?

- A 3
- B 4
- C 5
- D 8

6 UNIT 3 • SINUSOIDAL MODEL

A rider's height is $h(t) = 10 - 8 \cos(\pi t/12)$ meters, t in minutes. When is the rider **first** 14 m high?

- A 4 min
- B 6 min
- C 8 min
- D 12 min

7 UNIT 3 • TRIG EQUATIONS

How many solutions does $2 \sin^2 \theta - \sin \theta - 1 = 0$ have on $0 \leq \theta < 2\pi$?

- A 1
- B 2
- C 3
- D 4

8 UNIT 3 • POLAR

For $r = 2 - 4 \cos \theta$ on $0 \leq \theta < 2\pi$, for which values of θ is r **negative**?

- A $\pi/3 < \theta < 5\pi/3$
- B $0 \leq \theta < \pi/6$ or $11\pi/6 < \theta < 2\pi$
- C r is never negative
- D $0 \leq \theta < \pi/3$ or $5\pi/3 < \theta < 2\pi$

9 UNIT 1 • RATES OF CHANGE

A function has values $f(0) = 2$, $f(1) = 5$, $f(2) = 9$, $f(3) = 14$, $f(4) = 20$. Which model fits these values exactly?

- A Linear
- B Exponential
- C Quadratic
- D Logarithmic

10 UNIT 2 • COMPOSITION

$f(x) = \sqrt{x - 1}$ and $g(x) = x^2 + 1$. Which describes $g(f(x))$?

- A x for all real x
- B x for $x \geq 1$
- C x^2 for $x \geq 1$
- D $\sqrt{x^2}$ for all real x

09

CHECK YOUR WORK

Answer key & explanations

Read every explanation, even for questions you got right. Each one points to a pattern that keeps coming back on the exam.

#	Answer	Why
1	A	$f = (x - 2)(x + 2)/((x - 2)(x + 1))$. The $x - 2$ cancels (hole at $x = 2$); $x + 1$ stays (VA at $x = -1$). Equal degrees give HA $y = 1$.
2	D • (-1, 0) only	Leading term $-2x^6 \rightarrow$ negative at both ends. Signs change at -1 (odd) and 0 (odd) but not at 3 (even). Testing gives $+$ only between -1 and 0 .
3	B • 8.4%	Yearly factor $= 1.5^{1/5} \approx 1.084$, so about 8.4% per year. 50% is the growth per 5 years.
4	A • {8}	Combine: $\log_2(x(x - 6)) = 4$, so $x^2 - 6x - 16 = 0 \rightarrow x = 8$ or -2 . $x = -2$ makes the logs undefined, so it is extraneous.
5	B • 4	Solve $f(x) = 28$: $3 \cdot 2^{x-1} = 24 \rightarrow 2^{x-1} = 8 \rightarrow x - 1 = 3 \rightarrow x = 4$.
6	C • 8 min	$\cos(\pi t/12) = -1/2$ first when $\pi t/12 = 2\pi/3$, so $t = 8$. Choice A uses $+1/2$, forgetting that the rider starts at the minimum.
7	C • 3	Factor: $(2 \sin \theta + 1)(\sin \theta - 1) = 0$. $\sin \theta = 1$ gives $\pi/2$; $\sin \theta = -1/2$ gives $7\pi/6$ and $11\pi/6$.
8	D	$r < 0$ when $\cos \theta > 1/2$, which is near $\theta = 0$: $0 \leq \theta < \pi/3$ or $5\pi/3 < \theta < 2\pi$.
9	C • Quadratic	First differences 3, 4, 5, 6 are not constant, but second differences are all 1, so a quadratic fits exactly.
10	B	$g(f(x)) = (\sqrt{x - 1})^2 + 1 = x$, but only where f is defined: $x \geq 1$. The domain of the inner function carries through.

9–10

Strong. Move on to full timed free-response practice.

6–8

On track. Revisit the unit behind every miss on page 4.

0–5

Build foundations now. Re-learn rates of change and factoring (Unit 1), then logs and the unit circle.

10 KNOW THE TARGET How students score

In recent years, about 81–82% of students earned a 3 or higher on AP Precalculus, and about 28–29% earned a 5. Precalculus has one of the higher 3+ rates, so aim for a 4 or 5.

81–82%

of students earned a 3 or higher in recent years

28–29%

earned the top score of 5 in recent years

3+

is the score many colleges accept for credit or placement

What each AP score means

5

Extremely well qualified

4

Very well qualified

3

Qualified

2

Possibly qualified

1

No recommendation

Ranges reflect recent College Board score distributions and shift a little each year. Credit and placement policies vary by college, so check each college's AP policy.

Exam-day checklist

- ☐ **Bluebook** is installed and updated, and your device passed the exam check
- ☐ Device **fully charged**, plus **two pencils and a dark-ink pen** for the free-response booklet
- ☐ Admission ticket and photo ID, as your school requires
- ☐ Confirm your **exam day and start time** in May with your AP coordinator
- ☐ An **approved graphing calculator** with fresh batteries, set to **radians**
- ☐ Plan your time: about **2¼ min** per Part A question and **17 min** per free-response question
- ☐ Answer **every** multiple-choice question. Wrong answers don't lose points
- ☐ Keep answers **exact** in no-calculator parts

In the free-response section: label each part, write models with real numbers, interpret every parameter in context with units, and show each algebra step in Part B.



ACT SAT GO • 1-ON-1 AP TUTORING

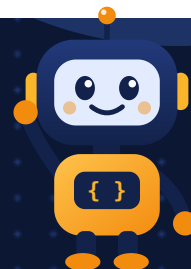
Aiming for a 5? Get a personal AP Precalc plan.

Book a free diagnostic lesson. We'll find your weakest units, review your practice responses, and give you a week-by-week plan to exam day in May.

Book a free diagnostic lesson →



Scan to book
www.actsatgo.com/consultation



Concept tutoring

Live 1-on-1 sessions on all three tested units, with no-calculator drills after every lesson.

Free-response coaching

Practice modeling and symbolic questions, marked against rubric-style criteria.

Timed exam practice

Full-length multiple-choice sets with score reports, so you see progress before May.

◆ Contact us

Web www.actsatgo.com

Email info@actsatgo.com

Phone [+1 \(945\) 391-6179](tel:+19453916179)

Share this guide

Know a classmate taking AP Precalculus? Send them the link. It's free to download at www.actsatgo.com, along with guides for every other AP subject.

AP® and Advanced Placement® are trademarks registered by College Board, which is not affiliated with, and does not endorse, this product. Exam structure, weightings, question types and score information are summarized from publicly available College Board materials. Practice questions, sample responses and study plans are original works by ACT SAT GO and are not official exam questions or scoring guidelines. © ACT SAT GO. Free to share for personal, non-commercial study use.